Running Claude and Local Models with Ollama

Keep working when you hit your Claude usage limit, or if you do not have a paid plan. Ollama runs open-weight models on your own laptop, free and offline, with no Anthropic account. This guide gives you three routes, from simplest to most capable.

Which route should I use?

Route	Terminal needed?	What you get
A. Ollama app chat window	No	A simple chat window. The quickest start. Best for questions, explanations and quick code snippets.
B. Open WebUI in your browser	Only to install and start it	A ChatGPT-style chat page with conversation history, model switching and file uploads.
C. Claude Code on a local model	Yes	An agent that reads your files and runs commands, powered by a local model instead of Claude. Weaker, but works for simple tasks.

All three start with the same two steps: install Ollama and download a model.

> **Be realistic.** Local models are much less capable than Claude, and slower on a laptop. They are good for quick plots, exploration and simple coding. Use Claude for complex reasoning and long analyses while you still have usage.

1. Install Ollama

Ollama is a free app that downloads and runs open-weight models. It works on macOS and Windows.

macOS

1. Go to **ollama.com/download**, download the macOS installer, and drag Ollama into your Applications folder.
2. Open Ollama from Applications. It runs quietly in the menu bar.

Windows

1. Go to **ollama.com/download**, download the Windows installer and run it.
2. Ollama starts automatically and lives in the system tray.

Check it worked by opening Terminal (macOS) or PowerShell (Windows) and running:

```
ollama --version
```

2. Choose and download a model

Pick by how much memory (RAM) your laptop has. Bigger models are smarter but slower. You can keep several installed.

Your RAM	Model	Download (approx.)	Good for
Under 8 GB	`llama3.2:3b` (or `llama3.2:1b` on the slowest machines)	2.0 GB (1.3 GB)	Quick demos and simple questions. Runs on modest laptops.
8 GB	`qwen3:8b` (recommended start)	~5 GB	Reasoning, code and data analysis. Close other apps while it runs.
16 GB	`qwen3:14b`	~9 GB	Noticeably better at multi-step reasoning.
32 GB	`qwen3:32b`	~19 GB	Strongest general option for most laptops.
16 GB or more, for code	`devstral:24b`	~14 GB	Mistral's coding model. Good at Python.

Sizes are approximate. Newer model versions appear regularly, so check the full catalogue at **ollama.com/library** for current names and sizes.

Download your chosen model (replace the name if you picked another):

```
ollama pull qwen3:8b
```

Quick test. A sensible answer means Ollama works. Press `Ctrl+D` to exit.

```
ollama run qwen3:8b "What causes El Niño?"
```

3. Route A: the Ollama app chat window

Recent versions of the Ollama app include a built-in chat window, so you do not need the terminal or a browser.

1. Open the Ollama app from the menu bar (macOS) or system tray (Windows).
2. Choose the chat window, pick your downloaded model from the model menu, and type your question.
3. You can drag text or PDF files into the chat. If answers seem to forget earlier messages, raise the context length in the app's settings (this uses more memory).

If you cannot see a chat window, update Ollama from ollama.com/download, or use Route B.

4. Route B: Open WebUI, a chat page in your browser

Open WebUI connects to your local Ollama models and gives you conversation history, model switching and file uploads. After the one-time install you work in the browser.

> **Python version.** Open WebUI needs **Python 3.11 or 3.12**. Python 3.13 is not supported yet. Check yours with `python --version`. If you are on 3.13, use the conda option below.

Install (once)

Create a clean environment and install. Choose one option.

Option 1: conda (recommended if you set up the course environment)

```
conda create -n webui python=3.11
conda activate webui
pip install open-webui
```

Option 2: Python venv

```
python3 -m venv webui-env
source webui-env/bin/activate        # Windows: webui-env\Scripts\activate
pip install open-webui
```

Start it (each time)

```
open-webui serve
```

1. Open your browser at **http://localhost:8080**.
2. Create an account. The first user becomes the admin. This account is local to your computer.
3. Choose your Ollama model from the dropdown and start chatting.

Make sure the Ollama app is running first. To stop the server, press `Ctrl+C` in the terminal where it is running.

5. Route C: Claude Code on a local model

Use this if you want the Claude Code experience (reading files, running commands) without an Anthropic account. It needs a recent Ollama (version 0.14 or later) and works best with a model that handles tool use, such as the ones in the table above.

One-time setup

macOS / Linux

```
curl -fsSL https://claude.ai/install.sh | bash     # installs Claude Code
```

Windows (PowerShell)

```
irm https://claude.ai/install.ps1 | iex
```

Then download a model (see section 2) and check both tools are installed:

```
ollama --version
claude --version
```

Start a session (every time)

Run this in one terminal window, from the folder you want to work in. Typing the three variables on the same command is what points Claude Code at your local model instead of Anthropic.

macOS / Linux

```
ANTHROPIC_AUTH_TOKEN=ollama ANTHROPIC_BASE_URL=http://localhost:11434 \
ANTHROPIC_API_KEY="" claude --model qwen3:8b
```

Windows (PowerShell)

```
$env:ANTHROPIC_AUTH_TOKEN="ollama"; $env:ANTHROPIC_BASE_URL="http://localhost:11434";
$env:ANTHROPIC_API_KEY=""
claude --model qwen3:8b
```

Swap `qwen3:8b` for any model you have pulled. Run `ollama list` to see the exact names.

What to expect

- A one-line warning that the model "isn't described by this version's model catalog". This is normal for non-Anthropic models and harmless.
- The first message is slow because the model is loading into memory. Later messages are much faster.
- A one-time "trust this folder?" prompt the first time you run Claude Code in a new folder. Accept it.
- Some Claude Code features, such as subagents, image input and complex tool use, may not work with local models. Keep tasks small and check the results.

6. A sensible workflow

- **While you have Claude usage:** use Claude for complex reasoning, long analyses and writing.
- **Quick plots, exploration and simple coding:** local models handle these well, so use them to save your usage.
- **Keep both ready.** Leave Ollama running, so when Claude says you have reached your limit, you can switch at once.

Troubleshooting

Symptom	Fix
`ollama: command not found`	Ollama is not installed, or the terminal needs restarting after install.
Open WebUI cannot see any models	Make sure the Ollama app is running (menu bar on macOS, system tray on Windows). Open WebUI looks for it at `http://localhost:11434`. Check you have pulled a model with `ollama list`.
`open-webui: command not found`	Activate the environment you installed it in (`conda activate webui` or the venv), then try again. Or run `python -m open_webui serve`.
Open WebUI install fails or stalls	It has many dependencies. Try again in a clean environment on Python 3.11. If it still fails, use Route A, or `ollama run qwen3:8b` in the terminal for a working chat.
Model runs very slowly	Try a smaller model (`qwen3:4b` or `llama3.2:3b`). Close memory-heavy apps such as many Chrome tabs or Docker. On a Mac, check Memory Pressure in Activity Monitor.
Claude Code: `Please run /login` or `401 Invalid bearer token`	The three `ANTHROPIC_...` variables were not set for that command. Retype the full line from "Start a session" in the same terminal window.
Claude Code: nothing happens for a long time	Normal on the first message. Wait 1 to 2 minutes for the model to load, especially larger ones.
Model name not recognised	Run `ollama list` and use the exact name shown, including the `:tag` suffix.

Help

Post in the course Slack channel, or open an Issue on the course GitHub repository (muschi-lab/MPhil-in-Past-and-Future-Climates-2026) and include the full error message and your operating system.

Sources: ollama.com/download, ollama.com/library, docs.ollama.com/api/anthropic-compatibility, docs.openwebui.com. Instructions checked on 12 Oct 2026 and may change as the apps update.